



Tobias Stål

GEOSCIENTIST AND ENGINEER

Tasmania, Australia

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Geophysicist working on Antarctica's ice-covered interior. I integrate geology, geophysics and statistical modelling to understand interactions between the ice sheet and the solid Earth, with a focus on geothermal heat, subglacial geology and instrumentation. Background in geology and geomorphology from the Arctic and Europe, and an earlier professional career in sound, lighting and electrical engineering.

Academic positions

- 2022 – Present **Research Associate in Computational Physics (Geophysics)**, ACEAS, University of Tasmania, Tasmania, Australia
- 2021 – 2022 **Research Fellow in Antarctic Seismology**, University of Tasmania, Tasmania, Australia
- 2020 – 2021 **Research Assistant**, University of Tasmania, Tasmania, Australia
- 2016 – 2021 **Casual tutor, demonstrator and research assistant**, University of Tasmania, Tasmania, Australia

Education

- 2021 **PhD, Geophysics, Geology and Computational Methods**, University of Tasmania, Australia
Thesis: *The Antarctic lithosphere revealed by multivariate analysis*
Supervisors: AM Reading, JA Halpin, JM Whittaker
Coursework in statistics and computational geophysics
- 2015 **MSc, Geophysics**, University of Copenhagen, Denmark
Thesis: *High-resolution S- and P-wave seismic mapping of near-surface chalk deposits*
Supervisor: L Nielsen
Coursework in geophysics, stochastic modelling and sedimentary basins
- 2011 **BSc, Geology**, University of Copenhagen, Denmark
Including studies at UNIS, Svalbard
Thesis: *Processes and development of the Faroe North Slide Complex*
Supervisors: GK Pedersen, T Nielsen
Coursework in geology, palaeontology, geomorphology, Quaternary geology and Arctic marine geology
- 1998 **AD, Sound Engineering**, Piteå Music Conservatory (Luleå University of Technology), Piteå, Sweden
Coursework in electronics, acoustics, radio broadcasting, music theory and recording technology
Including exchange studies at VGIK, Moscow, Russia
- Ongoing **Unincorporated studies (162.5 ECTS)**
Linguistics (linguistic typology, sociolinguistics, African languages, Nordic studies, pragmatics, grammar)
Geology: climate change, non-Nordic geology (the American Cordillera), Arctic petroleum provinces, geohazards, GIS and remote sensing

Peer-reviewed publications

- 2026 JA Halpin, NR Daczko, JA Mulder, A Maritati, **T Stål** — Ediacaran–Cambrian High-Strain Melt-Present Deformation of the Archean Charcot Province, East Antarctica. *Tectonics*, 45(8), e2026TC009479. doi:10.1029/2026TC009479
- 2026 MC Manassero, K Selway, **T Stål**, M Scheiter, M Loesing, F McCormack, JA Halpin, B Kulesa, AM Reading — Bed Topography and Subglacial Conditions of Denman Glacier, East Antarctica: Insights From Magnetotelluric Data and Interdisciplinary Studies. *Journal of Geophysical Research: Earth Surface*, 131(7), e2026JF009277. doi:10.1029/2026JF009277
- 2026 ID Kelly, AM Reading, **T Stål**, B Kulesa, A García-Jerez, J Piña-Flores, E Paolucci, A Tanzini, RJ Turner, JC Magyar et al. — Determining the Character of Subglacial Sediments in the Ice-Bedrock Interface Zone of Antarctica Using Horizontal-to-Vertical Spectral Ratios (HVSRS) of Seismic Ambient Noise. *Journal of Geophysical Research: Solid Earth*, 131(6), e2025JB033029. doi:10.1029/2025JB033029
- 2026 FS McCormack, **T Stål**, N Shao, E MacKie, A Fabela Hinojosa, M Lösing, J Roberts, S Ehrenfeucht, C Dow — Synthetic bed topographies for Antarctica and their utility in ice sheet modelling. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 384(2319), 20240537. doi:10.1098/rsta.2024.0537
- 2026 D Arora, NC Pant, **T Stål**, A Naik, M Pandey, R Gupta — Geothermal heat flow in the Princess Elizabeth Land sector, East Antarctica. *Earth-Science Reviews*, 277, 105475. doi:10.1016/j.earscirev.2026.105475
- 2026 M Al-Aghbary, M Sobh, **T Stål**, MO Awaleh, M Jalludin, C Gerhards — Uncertainty quantification using

clustering-based quantile regression forests: with an application to improving geothermal heat flow predictions. *Geophysical Journal International*, 245(2), ggag113. doi:10.1093/gji/ggag113

- 2026 M Lösing, W Colgan, **T Stål**, J Ebbing, AG Busck, T Zhang, HL Seroussi, F McCormack, D Fahrner, L Stearns et al. — Community heat flow recommendations: suitable basal boundary conditions for Greenland and Antarctica in ISMIP7. *GEUS Bulletin*, 62. doi:10.34194/r0w9rf81
- 2025 M Lösing, A Aitken, J Ebbing, JA Halpin, L Li, M Moorkamp, A Reading, **T Stål** — Linking Tectonics and Crustal Thermal Properties in Southwestern Australia and East Antarctica Through Coupled Gravity and Magnetic Analysis. *Journal of Geophysical Research: Solid Earth*, 130(12), e2024JB030770. doi:10.1029/2024JB030770
- 2025 NJ Abram, A Purich, MH England, FS McCormack, JM Strugnell, DM Bergstrom, TR Vance, **T Stål**, B Wienecke, P Heil et al. — Emerging evidence of abrupt changes in the Antarctic environment. *Nature*, 644(8077), 621–633. doi:10.1038/s41586-025-09349-5
- 2024 **T Stål**, JA Halpin, JW Goodge, AM Reading — Geology Matters for Antarctic Geothermal Heat. *Geophysical Research Letters*, 51(13), e2024GL110098. doi:10.1029/2024GL110098
- 2022 FS McCormack, JL Roberts, CF Dow, **T Stål**, JA Halpin, AM Reading, MJ Siegert — Fine-Scale Geothermal Heat Flow in Antarctica Can Increase Simulated Subglacial Melt Estimates. *Geophysical Research Letters*, 49(15), e2022GL098539. doi:10.1029/2022GL098539
- 2022 **T Stål**, AM Reading, S Fuchs, JA Halpin, M Lösing, RJ Turner — Properties and biases of the global heat flow compilation. *Frontiers in Earth Science*, 10, 963525. doi:10.3389/feart.2022.963525
- 2022 AM Reading, **T Stål**, JA Halpin, M Lösing, J Ebbing, W Shen, FS McCormack, CS Siddoway, D Hasterok — Antarctic geothermal heat flow and its implications for tectonics and ice sheets. *Nature Reviews Earth & Environment*, 3(12), 814–831. doi:10.1038/s43017-022-00348-y
- 2022 PE Morse, AM Reading, **T Stål** — Exploratory Volumetric Deep Earth Visualization by 2.5D Interactive Compositing. *IEEE Transactions on Visualization and Computer Graphics*, 28(7), 2641–2653. doi:10.1109/TVCG.2020.3037226
- 2021 **T Stål**, AM Reading, JA Halpin, JM Whittaker — Antarctic Geothermal Heat Flow Model: Aq1. *Geochemistry, Geophysics, Geosystems*, 22(2), e2020GC009428. doi:10.1029/2020GC009428
- 2021 G Sanchez, JA Halpin, M Gard, D Hasterok, **T Stål**, T Raimondo, S Peters, A Burton-Johnson — PetroChron Antarctica: A Geological Database for Interdisciplinary Use. *Geochemistry, Geophysics, Geosystems*, 22(12), e2021GC010154. doi:10.1029/2021GC010154
- 2020 **T Stål**, AM Reading, JA Halpin, SJ Phipps, JM Whittaker — The Antarctic Crust and Upper Mantle: A Flexible 3D Model and Software Framework for Interdisciplinary Research. *Frontiers in Earth Science*, 8, 577502. doi:10.3389/feart.2020.577502
- 2020 **T Stål**, AM Reading — A Grid for Multidimensional and Multivariate Spatial Representation and Data Processing. *Journal of Open Research Software*, 8(1), 2. doi:10.5334/jors.287
- 2019 **T Stål**, AM Reading, JA Halpin, JM Whittaker — A Multivariate Approach for Mapping Lithospheric Domain Boundaries in East Antarctica. *Geophysical Research Letters*, 46(17-18), 10404–10416. doi:10.1029/2019GL083453
- 2019 PE Morse, AM Reading, **T Stål** — Well-Posed Geoscientific Visualization Through Interactive Color Mapping. *Frontiers in Earth Science*, 7, 274. doi:10.3389/feart.2019.00274

Submitted and in review

- 2026 MC Manassero, K Selway, AM Reading, N Askey-Doran, JR Peacock, F Ramirez, **T Stål** — Magnetotellurics, mantle viscosity, integrated geophysics, glacial isostatic adjustment, mantle water content, East Antarctica. *Journal of Geophysical Research: Solid Earth*. In review.

Outreach

- 2024 **T Stål**, FS McCormack, AM Reading, N Askey-Doran, JA Halpin, M Lösing — Geothermal Heat Shapes the Antarctic Ice Sheet From Below. *Frontiers for Young Minds*, 11, 1178537. doi:10.3389/frym.2023.1178537

Datasets

- 2024 Global Heat Flow Data Assessment Group, S Fuchs, F Neumann, B Norden, E Balkan-Pazvantoglu, S Elbarbary, A Petrulin, G Beardsmore, R Harris, ..., **T Stål** — The Global Heat Flow Database: Release 2024 [Dataset]. *GFZ Data Services*. doi:10.5880/fidgeo.2024.014
- 2024 AM Reading, **T Stål**, I Kelly — Ice Sheets and Interactions, Dronning Maud Land and Coats Land [Dataset]. *International Federation of Digital Seismograph Networks*. doi:10.7914/J9CQ-XD60
- 2023 Global Heat Flow Data Assessment Group, S Fuchs, F Neumann, B Norden, G Beardsmore, P Chiozzi, W Colgan, AP Anguiano Dominguez, MRA Duque, ..., **T Stål** et al. — The Global Heat Flow Database: Update 2023 [Dataset]. *GFZ Data Services*. doi:10.5880/fidgeo.2023.008
- 2023 **T Stål**, AM Reading, S Kupis — Ice Sheets and Interactions, East Antarctica [Dataset]. *International Federation of Digital Seismograph Networks*. doi:10.7914/74EN-YT90

- 2021 **T Stål**, AM Reading, S Kupis, J Newlands — Casey-Wilkins Adaptable Array [Dataset]. *International Federation of Digital Seismograph Networks*. doi:10.7914/SN/Z9_2021
- 2020 **T Stål**, AM Reading, JA Halpin, SJ Phipps, J Whittaker — The Antarctic crust and upper mantle: a flexible 3D model and framework for interdisciplinary research [Dataset]. *PANGAEA*. doi:10.1594/PANGAEA.918549
- 2015 AM Reading, **T Stål** — CAD Seismic Deployment, Casey-Davis Region, East Antarctica [Dataset]. *International Federation of Digital Seismograph Networks*. doi:10.7914/NRW9-9825

Software and code

- 2022 **T Stål** — Fast Calculation of Ripley's K and L Functions on a Sphere [Software]. *Zenodo*. doi:10.5281/zenodo.6865005
- 2019 **T Stål** — agrid: a grid for multidimensional and multivariate spatial modelling [Software]. *Zenodo*. doi:10.5281/zenodo.2553966

Selected conferences, workshops and hackathons

- 2026 EGU General Assembly, Vienna, Austria
– Aq2 and Kq2, refined geothermal heat flow models from multivariate observables for application to ice sheet modelling
- 2025 geoBRIDGE Working Group Hackathon, Perth, Australia
– Participant and project contributor
- 2025 EGU General Assembly, Vienna, Austria
– Seismicity of Denman Glacier: constraints on geometry and dynamics
- 2024 SCAR Open Science Conference, Pucón, Chile
– Aq2: a refined geothermal heat flow map of Antarctica from multivariate observables for application to ice sheet modelling
– Exploring Atlas Antarkitiki: a distinctive repository of historical Antarctic and Southern Ocean data
– Convened three scientific sessions on Antarctic geology and geophysics, and co-organised a side meeting to organise collaborations at Bunker Hills
- 2023 EGU General Assembly, Vienna, Austria
– Using information entropy to optimise and communicate certainty of continental scale tectonic models
- 2023 Maydena Subglacial Hydrology and Geology Workshop, Maydena, Australia
– The subglacial heat budget – a conceptual investigation
– Organised and co-chaired the workshop
- 2021 AGU Fall Meeting, New Orleans and online
– Modelling Antarctic subglacial geothermal heat transfer in higher resolution, with reduced uncertainty and assumptions addressed
– Co-convened a session on Antarctic geothermal heat
- 2021 Australian Earth Sciences Convention (AESC) "Core to Cosmos", online
– Data-driven tectonic regionalisation of Antarctica: appreciate the similarity
– Convened two sessions on data science and machine learning
- 2020 SCAR Open Science Conference: Antarctic – Global Connections, online
– Towards a reconciled heat flow map for Antarctica: Aq1
– The Wilkes Land sector including the Aurora Basin, and its most probable subglacial geology
- 2020 AGU Fall Meeting, online
– Multivariate geothermal heat flow models of Antarctica, towards Aq2
- 2019 Geological Society of Australia SGTSG Meeting, Port Lincoln, Australia
– The probable geology of the Aurora Basin, East Antarctica
- 2019 XIII International Symposium on Antarctic Earth Sciences (ISAES), Incheon, Korea
– Linking Antarctic geological observations and geophysical data in a probabilistic space
- 2018 POLAR2018 Open Science Conference, Davos, Switzerland
– A multi-domain lithospheric model of East Antarctica
- 2018 SCAR TACTical heat flow workshop, Hobart, Australia
– Talk and poster on the tectonic structure of Antarctica
- 2018 IEEE Antarctic and Southern Ocean Forum (ASOF), Hobart, Australia
– A frame for processing multivariate Antarctic solid Earth data
- 2018 DataTas seminar, Hobart, Australia
– Software construction tools make everything so easy!
- 2018 Agile* Subsurface Hackathon, Copenhagen, Denmark
– Winner of the People's Choice Award

2017	AGU Fall Meeting, New Orleans, USA – Combined constraints on the structure and physical properties of the East Antarctic lithosphere from geology and geophysics
2015	Workshop on 3D Reflection Seismic Processing Using Open-Source Software, Rice University, Houston, Texas, USA

Awards and recognition

2023	Royal Society of Tasmania Doctoral Award
2021	Wiley Top Cited Article
2020	PhD Award for Outstanding Performance during Postgraduate Studies
2019	Progress in Inclusion, Diversity & Equity

Memberships and service

2023 – Present	Member, International Heat Flow Commission (IHFC), IUGG
2021 – Present	Co-chair, SCAR INSTANT Heat Flow Subcommittee
2020	Organising committee, Australian Antarctic Research Conference (AARC)
2019	Organising committee, Geological Society of Australia SGTSG Meeting
2017 – 2020	Member, School of Natural Sciences Inclusion, Diversity & Equity Committee

Selected non-academic work experience

2021 – Present	GRIT Facility Manager – Instrumentation , University of Tasmania, Tasmania, Australia Instrumentation management and Antarctic fieldwork support
2019 – 2021	Geoscientist , Geoneon, Australia Marine geoscience, GIS and data compilation
2011 – 2012	Invited technician and video designer , Al-Harah Theatre, Palestine
2010 – 2015	Field Leader and Geophysicist , COWI, Denmark MEP, GPR and TEM fieldwork, and GIS-based survey design and interpretation
2010	Field Scientist , Norwegian Polar Institute, Svalbard Supporting biological fieldwork
2009 – 2016	Lighting Designer , Rapid Eye, Denmark Including the award-winning performance <i>QUIPROQUO</i>
2009 – 2014	Lighting Designer , STÜ, Estonia Including touring
2008 – 2009	Technical Producer , National Theatre of Ghana, Ghana
2005 – 2007	Sound/Video/Light Designer and Technician , The Swedish National Touring Theatre, Sweden And shorter projects until 2011
2004 – 2006	Sound Engineer , Circus Cirkör, Sweden and world tour
2003 – 2004	Sound Designer , The Royal Dramatic Theatre, Sweden
1998 – 2003	Stage Manager , Uppsala Stadsteater, Sweden
1997 – 2004	Sound Engineer , Swedish Radio (P1, P3), Sweden
1993 – 1996	Plumbing Apprentice , Linds Rör AB, Sweden

Art collaborations (as a scientist)

2025	T Stål, AM Reading, K Verell — <i>Seismic Dance Party</i> , Beaker Street (Hobart). Collaborative science-art performance with AuScope, sound designer K Verell and DJs Huh Pink, translating Denman Glacier seismic data into an immersive audio-visual dancefloor experience.
2025	L Bleach — <i>An Infrasonorous Archipelago</i> , Walk&Talk Biennial "Gestures of Abundance" (Azores). Using seismo-volcanic infrasound from the Azores to create tactile and sonic works that tune audiences to the Earth's low-frequency vibrations.
2025	M Nieboer — <i>Through Our Eyes</i> , Australian Embassy (Paris). Exhibition on women's experiences and narratives, the culmination of two years of work; I contributed cartographic design and geospatial visualisation.
2023	L Bleach — <i>soft rock monitoring (the slow seismogenic zone)</i> , EARTHEN. Better Nature, Cement Fondu (Sydney). Collaboration with Lucy Bleach and Geoscience Australia, transmitting vibrational signals from the SYDS soft-rock urban seismic monitoring site into the gallery architecture via transducer speakers.
2021	L Bleach — <i>Attenuated ground (the slow seismogenic zone)</i> , TarraWarra Biennial 2021: Slow Moving Waters (TarraWarra Museum of Art, Victoria). Integrating double bass, cast toffee, seismometer, tactile transducers and

slow-earthquake signals from the Japan Trench to render seismogenic motion as resonant sound and vibration.

Other skills and experience

Coding	Python (scikit-learn, OpenCV, TensorFlow, dask, xarray, geopandas), GIS (QGIS, ArcGIS, GDAL/OGR), TeX/LaTeX (TikZ), Bash, Linux, macOS, git, SCons, Madagascar, Seismic Unix, SQL and MAX. Some HTML, C and JavaScript (learning). Microsoft Office, Adobe Creative Suite, Blender (BlenderGIS), GIMP, Inkscape, CAD (Fusion). Working but outdated knowledge of MATLAB, SAC, Emacs Lisp and BASIC.
Certificates	Coastal Skipper Certificate (SWE); Marine HF/VHF Radio Certificate (SWE); firearms licence and hunting certificate (SWE); PADI Advanced Open Water Diver (including dry suit and Nitrox); Wilderness First Aid (AUS); Australian Approved Arrangements Accreditation Biosecurity (AUS); trip leader and terrain-vehicle driver for the Australian Antarctic Division (e.g. Casey 2023)
Languages	English (C2), Swedish (native), Danish (fluent), French (intermediate), Russian (functional), Asante Twi (functional), Norwegian (reading and understanding), Esperanto (inactive), Spanish (learning), Yiddish (learning), Estonian (learning)
Online courses	Drawing Nature, Science and Culture: Natural History Illustration NHI101x (University of Newcastle, 2020); 3D Earth spring school (Kiel University, 2021); Presenting Data and Information by Edward Tufte (2025)
Practical skills	Welding, woodwork, diesel engines, sewing, cooking and baking (enquire about my chocolate mousse)
Other interests	Music, literature and theatre. I also love old boats, particularly tugs.